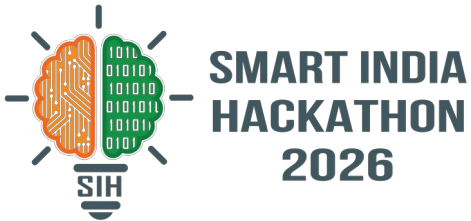


SMART INDIA HACKATHON 2026



RASTA AI **SIH26002**

SHORTEST IS NOT ALWAYS
USABLE.

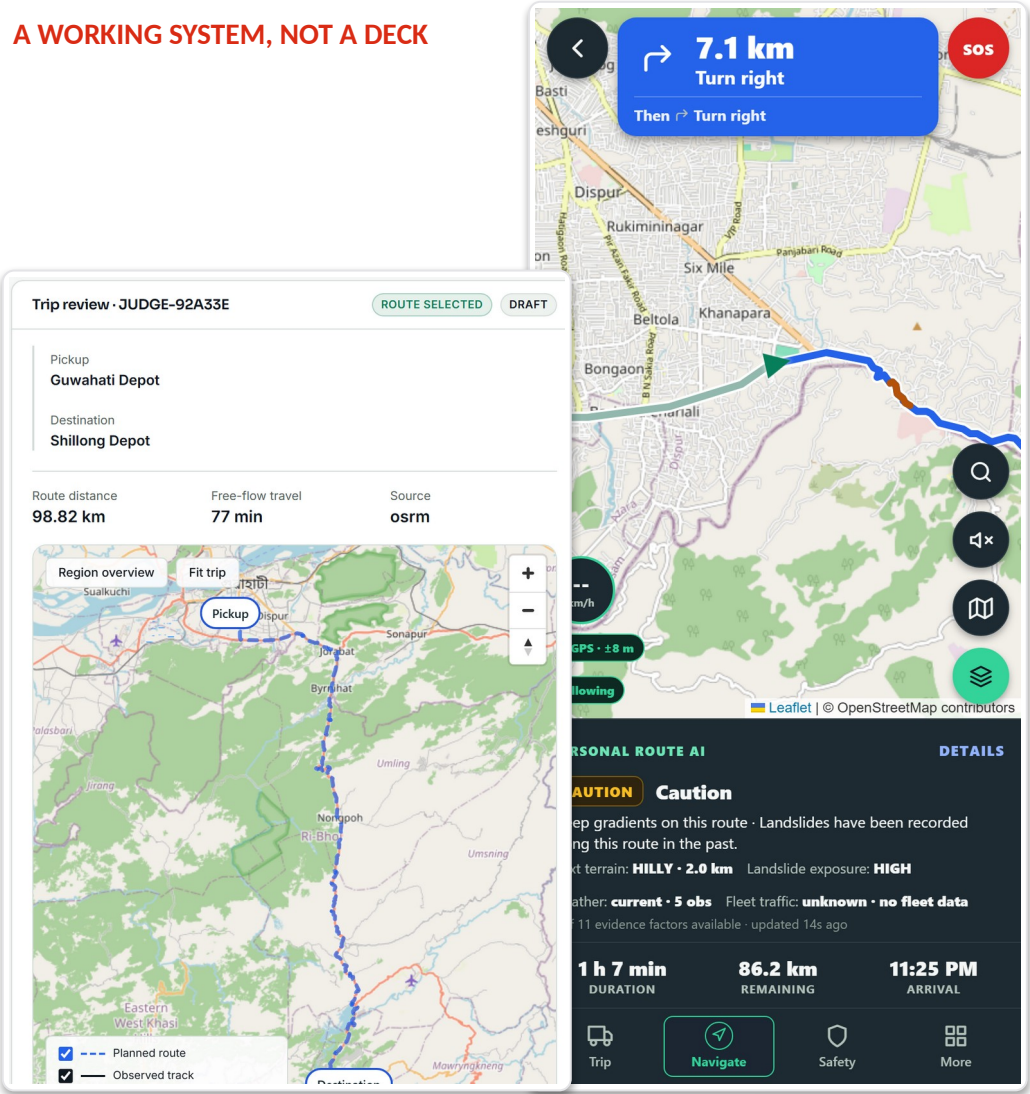
“Can this essential-supply truck reliably use this corridor now?”

PROBLEM STATEMENT	SIH26002 · MDoNER
TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform (NER)
THEME / CATEGORY	Smart Automation · Software
TEAM	NER-AI LOGISTICS · Team ID 17 (internal group no.)

Route Intelligence for Essential-Supply Logistics in North Eastern India

End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong

A WORKING SYSTEM, NOT A DECK



A road can be the shortest and still be **unusable**.

Monsoon · landslides · floods · official warnings · single-lane hill roads · no signal. **A shortest-path router still sends the truck.**

A NORMAL ROUTER

Origin → Shortest / fastest road → Truck

No governance. No answer when data is missing.

RASTA AI



Manager governance → driver navigation, 23 languages, offline package

UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN — never safe — and goes to an authorised reviewer.

MANAGER · CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN **0.5 KM STEEP**

49-1436 m · 1975 m climb · steepest 10.3%

flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km

Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY **HIGH EXPOSURE**

22 recorded landslides within 5 km of the road, nearest 0.4 km

6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two

NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS **NORMAL**

Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS **NONE ON CORRIDOR**

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC **UNKNOWN**

Real screen · evidence · freshness · UNKNOWN

Evidence decides. AI explains, and never overrules.

WEATHER Open-Meteo · MET Norway
TERRAIN Copernicus DEM · OpenTopoData
WARNINGS NDMA SACHET (CAP)
FLOOD GloFAS
LANDSLIDE HISTORY NASA GLC 2007–17
TRAFFIC fleet GPS fixes

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE**CAUTION****HOLD****REROUTE**

MANAGER REVIEW

DRIVER NAVIGATION

ROUTE-AHEAD 60 s

CONDITIONS CHANGE?

REASSESS

CLIENTS Manager console (React + TS) · Driver app (React Native / Expo, Android) · one login, server-decided role, offline package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

WHO DECIDES

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence



Deterministic safety policy



Manager governance



Driver action

It already runs on a real phone, on a real road.

Run end to end on a physical Android phone, hosted backend, real road.

12/12 physical judge-flow steps

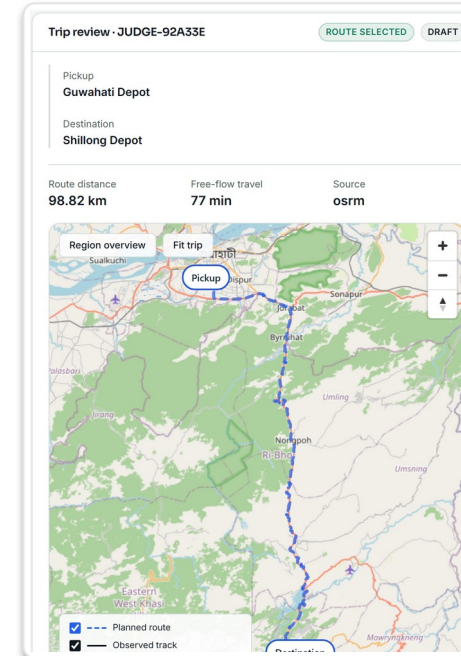
9/9 physical role-flow steps

1138+ backend · 621+ driver · 170+ manager tests

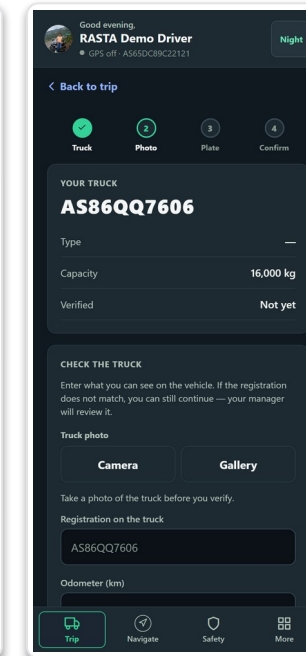
PHYSICAL ANDROID · CERTIFIED

Canonical demo Guwahati → Shillong · ≈ 98.8 km real route

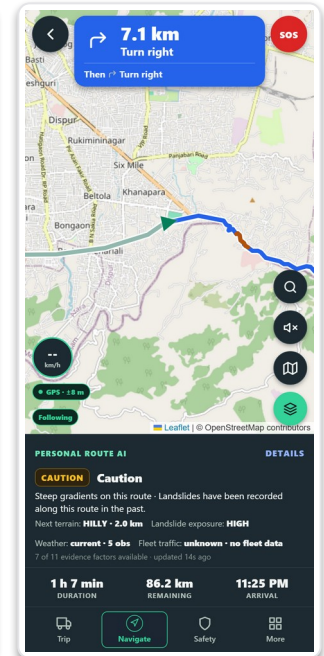
CERTIFIED LIFECYCLE



Manager · route review



Driver · truck check



Driver · navigation

KNOWN LIMITS → MITIGATION Cold start → warm /health first · rate limits → cached evidence + explicit UNKNOWN · push needs Firebase → in-app alerts work · experimental ML → outside routing

IMPACT AND BENEFITS

Food, medicine, fuel and relief reach people when the corridor is decided on evidence.

WHO BENEFITS

FLEET MANAGERS

Usability and uncertainty visible before dispatch. Auditable.

DRIVERS

Danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

Resilient access to food, medicine, fuel, relief.

GOVERNMENT / OPS

Auditable evidence and disruption per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

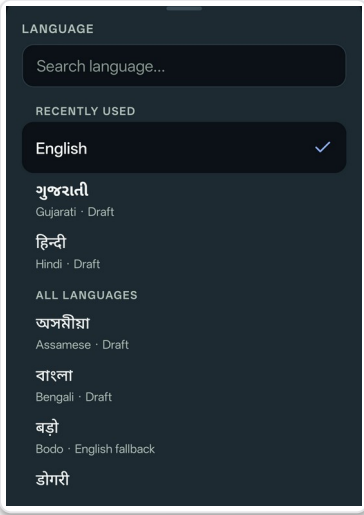
12/12 judge-flow steps certified on a physical phone

100 % of reroutes reviewed by a human

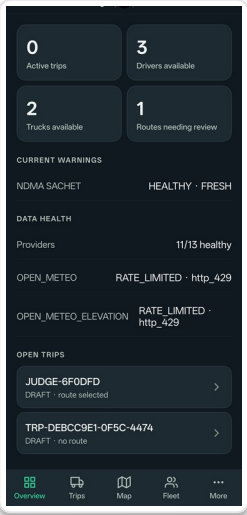
0 unknown factors counted as safe

Verified evidence → deterministic safety policy → manager governance → driver action. AI has no uncontrolled authority; no invented percentages.

REAL SCREENS • APK 1.0.18



Driver • 23 languages



Manager • mobile

APK 1.0.18. One login, server-decided role. Every language shows its status — Verified, Draft or English fallback.

We trained it, tested it, and **refused to ship it.**

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim
Road geometry, alternatives, turn steps.

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS
Official warnings with freshness; discharge vs 30-day mean.

AI ASSISTANCE

Google Gemini
Explains in the driver's language. Never safety authority.

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM •
OpenTopoData
Forecast along the road; climb, gradient, steep segments.

LANDSLIDE HISTORY

NASA Global Landslide Catalog
Historical exposure, 2007–2017, ≤ 5 km accuracy.

PROBLEM STATEMENT

MDoNER • SIH26002
Smart logistics and accessibility intelligence for the NER.

EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

NER geographic holdout • 154 events • no NER
row in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Recall was not enough. Held-out
testing exposed the false-positive rate. The gate keeps it out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 — calibrated logistic
regression, NASA GLC + ERA5-Land rain + DEM slope